INTELLIGENT AIR QUALITY PREDICTION AND ENVIRONMENTAL MONITORING SYSTEM

A Leak-Free Machine Learning Framework with Regularized Linear Classifiers and In-Fold Resampling for Chemical Multisensor Arrays

Project Code: AI-ML-060

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SECTION 2: DECLARATION AND ACKNOWLEDGEMENT

2.1 Declaration of Authorship

I hereby declare that this technical internship project report titled 'Intelligent Air Quality Prediction and Environmental Monitoring System' (Project Identification Code: AI-ML-060), submitted to the Department of Computer Science and Engineering, is an authentic record of original engineering and scientific work carried out by me. All data preprocessing workflows, mathematical feature derivations, machine learning pipeline architectures, cross-validation protocols, and web dashboard components presented in this document were implemented and verified inside the workspace environment.

I further affirm that this report adheres strictly to ethical standards and academic integrity. No dataset records, experimental metrics, confusion matrix values, or literature citations have been fabricated, contorted, or synthetically altered. The empirical findings reported herein represent genuine, unpadded results obtained from running reproducible Python scripts on the real-world benchmark atmospheric dataset. Where prior published literature or third-party open-source libraries have contributed to the methodology, explicit citations and references have been incorporated.

2.2 Acknowledgements

I wish to express my deepest and most sincere gratitude to the internship coordinators, senior machine learning engineers, and technical mentors whose invaluable insights, methodological rigor, and constructive evaluations guided the successful execution of this environmental machine learning prototype. Their dedication to leak-free data hygiene, robust statistical validation, and production-grade software engineering provided the foundation for this comprehensive investigation.

Special acknowledgement is extended to the University of California, Irvine (UCI) Machine Learning Repository and the Italian National Research Council (CNR-ISAC) for providing open access to the field multisensor atmospheric air quality dataset. The availability of high-resolution ground-truth chemical observations from co-located certified reference analyzers was indispensable in establishing the empirical baseline for this research.

Finally, I extend my heartfelt thanks to my family, academic peers, and colleagues for their constant encouragement, moral support, and patience throughout the conceptualization, model training, and documentation phases of this project.

SECTION 3: ABSTRACT

Atmospheric air pollution is recognized by the World Health Organization (WHO) as one of the leading global environmental risks to public health, contributing to millions of premature deaths annually through cardiorespiratory morbidity. Among toxic combustion byproducts, carbon monoxide (CO) represents an insidious threat due to its non-irritating, odorless, and asphyxiating nature. While municipal regulatory agencies employ certified reference analyzers (such as non-dispersive infrared photometers) to ensure precision, their substantial procurement expense, maintenance overhead, and physical bulk preclude high-density urban spatial coverage. Solid-state chemical multisensor arrays provide an economically scalable alternative for ubiquitous localized monitoring; however, their practical utility is severely impeded by cross-sensitivity to interfering gases, thermal drift, moisture-induced impedance shifts, and non-linear sensor response characteristics.

In this project (AI-ML-060), we architect, implement, and benchmark an end-to-end Intelligent Air Quality Prediction and Environmental Monitoring System designed to detect high-level carbon monoxide pollution events from multisensor responses. Using the official UCI Air Quality dataset (comprising 9,357 hourly field recordings from an Italian urban station across March 2004 to April 2005), we formulate a binary classification problem: predicting whether ambient carbon monoxide concentration exceeds the clinically critical hazard threshold of 4.0 mg/m³ (co_high = 1 if CO(GT) > 4.0 else 0). Crucially, to prevent target leakage, the ground-truth analyzer measurement CO(GT) is strictly excluded from all predictor feature vectors.

Our pipeline incorporates domain-specific feature engineering, including a sensor-to-sensor ratio (PT08.S1(CO) / (PT08.S3(NOx) + eps)), temperature-humidity interactions, and cyclic temporal coordinates. To combat severe class imbalance (10.58% minority high-CO instances), we enforce a mathematically leak-free resampling architecture using imbalanced-learn pipelines: missing values are imputed via median statistics; outliers are clipped using training-fitted Interquartile Range (IQR) thresholds; features are scaled; and Synthetic Minority Over-sampling Technique (SMOTE) is applied strictly within the training fold of a 5-fold Stratified Cross-Validation protocol.

We implement and comparatively evaluate two primary regularized linear paradigms: Model A (Lasso-style L1 regularized Logistic Regression, C=100.0, liblinear solver) and Model B (Ridge-

style L2 regularized Logistic Regression, C=1.0, lbfgs solver). On the untouched 20% held-out test set (N = 1,535), Model A achieved an exceptional accuracy of 94.59% (surpassing the 85.0% target benchmark), a weighted F1-score of 0.9498, a positive class recall of 94.44%, and an Area Under the ROC Curve (ROC-AUC) of 0.9873. The trained models are deployed within a full-stack Flask web application providing single-sample risk inference, batch CSV prediction, analytical reporting (TXT, HTML, DOCX, PDF), and comprehensive test coverage via pytest.

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SECTION 5: LIST OF ABBREVIATIONS, FIGURES, AND TABLES

5.1 List of Abbreviations

Table 5.1: List of Abbreviations

Abbreviation	Full Definition
AI	Artificial Intelligence
ML	Machine Learning
CO	Carbon Monoxide
CO(GT)	Carbon Monoxide Ground Truth (Certified NDIR Reference Analyzer)
MOS	Metal-Oxide Semiconductor
NDIR	Non-Dispersive Infrared Spectrophotometry
NMHC	Non-Methane Hydrocarbons
VOC	Volatile Organic Compounds
NOx	Nitrogen Oxides (Nitric Oxide and Nitrogen Dioxide)
NO2	Nitrogen Dioxide
O3	Ozone
T	Ambient Temperature (degrees Celsius)
RH	Relative Humidity (percentage)
AH	Absolute Humidity (kg/m3)
IQR	Interquartile Range
SMOTE	Synthetic Minority Over-sampling Technique
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve
CV	Cross-Validation
L1	Lasso (L1-norm) Regularization
L2	Ridge (squared L2-norm) Regularization
CNR-ISAC	National Research Council - Institute of Atmospheric Sciences and Climate (Italy)
WHO	World Health Organization
EPA	Environmental Protection Agency

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SECTION 6: INTRODUCTION

6.1 Environmental and Public Health Context of Air Pollution

Ambient air contamination has emerged as one of the defining epidemiological and ecological crises of the modern era. Uncontrolled urbanization, rapid vehicular fleet expansion, and localized industrial activity discharge an intricate cocktail of hazardous particulate matter and gaseous effluents into the troposphere. According to epidemiological assessments published by the World Health Organization (WHO) and the United Nations Environment Programme (UNEP), exposure to polluted ambient air contributes to more than seven million premature deaths globally every year. The adverse physiological consequences of chronic and acute exposure span ischemic heart disease, cerebrovascular accidents, chronic obstructive pulmonary disease (COPD), and exacerbated pediatric asthma. Consequently, municipal governments and environmental protection authorities require continuous, reliable, and actionable monitoring systems capable of identifying hazardous atmospheric contamination events.

6.2 Carbon Monoxide: Toxicology and Atmospheric Dynamics

Among the criteria atmospheric pollutants regulated under clean air directives, carbon monoxide (CO) occupies a uniquely lethal position. Generated predominantly by the incomplete combustion of carbonaceous hydrocarbon fuels in internal combustion automobile engines, heating furnaces, and industrial boilers, carbon monoxide is a colorless, odorless, and non-irritating gas. Unlike sensory-detectable pollutants such as sulfur dioxide or ozone, high ambient concentrations of carbon monoxide can accumulate without human sensory perception.

The toxicology of carbon monoxide stems from its extraordinarily high affinity for human hemoglobin—approximately 200 to 250 times greater than that of oxygen. Upon inhalation, CO molecules rapidly bind with hemoglobin to form carboxyhemoglobin (COHb), drastically impairing the oxygen-carrying capacity of systemic circulation and shifting the oxyhemoglobin dissociation curve to the left. The resultant systemic tissue hypoxia damages the central nervous system and myocardium, leading to dizziness, cognitive impairment, myocardial infarction, and death at elevated exposures. Ambient background concentrations typically hover between 0.5 and 2.0 mg/m^3, but heavily congested urban thoroughfares and street canyons frequently

experience acute spikes exceeding 4.0 mg/m³, presenting severe acute hazards to commuters, pedestrians, and nearby residents.

6.3 Low-Cost Chemical Sensor Networks and Challenges

Historically, atmospheric pollution monitoring has relied on stationary regulatory monitoring stations operated by government environmental agencies. These reference installations utilize certified analytical instruments, such as non-dispersive infrared (NDIR) photometers for carbon monoxide and chemiluminescence analyzers for nitrogen oxides. While reference instruments exhibit sub-parts-per-billion precision, they are constrained by high procurement expenditures (exceeding $30,000 to $50,000 per station), heavy power requirements, and large physical footprints. Consequently, municipal monitoring networks remain sparse, often featuring fewer than five reference stations across an entire metropolitan region, which obscures microclimatic pollution hotspots.

To overcome spatial sparsity, low-cost solid-state chemical sensors—specifically metal-oxide semiconductor (MOS) devices—have gained widespread traction. In a MOS sensor, a heated polycrystalline metal oxide film (such as tin oxide, titanium dioxide, or indium oxide) interacts with reducing or oxidizing atmospheric gases, altering electrical resistance. While MOS sensors are compact, energy-efficient, and economical, their raw electrical conductivities exhibit severe non-linear cross-sensitivity. A tin oxide CO sensor (e.g., PT08.S1) responds not only to carbon monoxide, but also to ambient temperature fluctuations, humidity shifts, and co-occurring volatile hydrocarbons. Accurately decoupling these confounding environmental interactions requires advanced statistical modeling.

6.4 Motivation for Machine Learning and Predictive Analytics

Machine learning techniques provide a transformative paradigm for chemical sensor signal processing. By modeling the intricate multi-dimensional interactions among multiple co-located solid-state sensors, ambient temperature, and humidity, machine learning algorithms can reconstruct ground-truth pollutant concentrations and predict hazardous atmospheric exceedances. Rather than attempting delicate continuous regression under severe drift, formulating the objective as an early-warning risk classification system (identifying whether CO

exceeds 4.0 mg/m³) provides municipal decision-makers with a robust, noise-resilient alert mechanism.

However, applying machine learning to real-world sensor datasets demands stringent methodological discipline. Naive data workflows often suffer from two catastrophic flaws: target leakage (inadvertently providing the model with features that contain the target label) and validation leakage (applying oversampling or scaling techniques prior to cross-validation splits). In this project, we engineer a mathematically rigorous, leak-free machine learning architecture that enforces strict data isolation while maximizing predictive performance.

SECTION 7: PROBLEM STATEMENT

7.1 Sensor Cross-Sensitivity and Drift Anomalies

The central challenge in low-cost environmental monitoring is the inherent unreliability of raw solid-state chemical sensor responses when deployed in unconditioned field environments. Metal-oxide semiconductor sensors operate through surface chemisorption phenomena, where ambient oxygen anions pre-adsorbed on the metal oxide surface react with reducing target gases. This chemical reaction is fundamentally sensitive to ambient temperature and water vapor concentrations, causing substantial baseline drift and temperature-induced sensitivity modulation. Furthermore, broad chemical cross-sensitivity means that a sensor designed for carbon monoxide also reacts to ambient nitrogen oxides, benzene, and non-methane hydrocarbons. Without sophisticated multivariate compensation, raw sensor readings cannot be directly trusted for environmental compliance or health hazard detection.

7.2 The Challenge of Ground-Truth Target Leakage

In academic and commercial predictive systems, a subtle yet pervasive flaw is target leakage. In the benchmark UCI Air Quality dataset, the target column to be predicted is whether the reference certified carbon monoxide analyzer reading, denoted as CO(GT), exceeds 4.0 mg/m^3. In unprincipled implementations, engineers frequently include CO(GT) or derived collinear metrics directly within the predictor matrix, resulting in near-perfect artificial accuracy (100% or 99.9%) during testing that completely fails in real-world deployment where reference analyzers are absent. To preserve practical utility and scientific validity, the model must predict hazardous carbon monoxide conditions solely from low-cost multisensor array responses (PT08.S1 through PT08.S5) and meteorological variables, with CO(GT) strictly excised from predictor inputs.

7.3 Severe Class Imbalance in Hazardous Pollution Events

In representative urban atmospheric records, hazardous pollution events are relatively infrequent compared to moderate or clean baseline conditions. In the UCI Air Quality dataset, hours where CO(GT) exceeds 4.0 mg/m^3 account for only 812 out of 7,674 valid labeled observations—a minority class prevalence of just 10.58% (an imbalance ratio of approximately 1 : 8.5). Standard

linear classifiers trained directly on such imbalanced distributions tend to collapse toward the majority class, achieving misleadingly high accuracy (~89.4%) by predicting 'Low CO' for all observations while completely failing to detect hazardous episodes (0% recall). Applying naive oversampling techniques like SMOTE on the whole dataset introduces severe data contamination by synthesizing training points informed by validation samples. Designing a pipeline that resolves severe class imbalance without compromising cross-validation integrity constitutes a critical objective.

SECTION 8: EXISTING SYSTEM

8.1 Traditional Reference Monitoring Stations

The incumbent paradigm for urban air quality surveillance relies almost entirely on automated urban air monitoring stations managed by environmental protection ministries. These stations house reference instruments including non-dispersive infrared (NDIR) carbon monoxide analyzers, chemiluminescent oxides of nitrogen analyzers, and ultraviolet photometric ozone analyzers. While these systems provide reliable measurements adhering to ISO/IEC standards, their physical dimensions and operating costs restrict deployments to a handful of fixed geographic coordinates. As a result, atmospheric variability across urban street canyons, school perimeters, and industrial corridors remains completely unmonitored.

8.2 Static Threshold Heuristics and Calibrated Linear Regression

Where low-cost sensors have been deployed in existing commercial prototypes, manufacturers traditionally apply static lookup tables, threshold-based heuristics, or unregularized Ordinary Least Squares (OLS) bivariate calibrations. These simplistic calibrations assume a linear relationship between sensor resistance and target gas concentration, ignoring thermodynamic moisture cross-sensitivities and multi-gas interference. When sudden meteorological shifts occur (such as sudden thermal inversions or humidity surges), unregularized calibrations generate massive false alarm rates or fail to detect genuine pollution episodes.

8.3 Limitations of Conventional Methodologies

A rigorous analysis of existing systems reveals four fundamental limitations:

• 1. Capital Expense and Spatial Sparsity: Reference stations cost tens of thousands of dollars to establish and maintain, resulting in sparse geographical coverage that fails to resolve microclimatic pollution hotspots.

• 2. Uncompensated Multi-Gas Interference: Existing heuristic threshold alarms on raw sensors fail to account for the simultaneous presence of NOx, benzene, and volatile hydrocarbons, resulting in frequent false positives during cold weather or high humidity.

• 3. Methodological Flaws in Prior ML Prototypes: Published literature frequently reveals inadvertent data leakage, where oversampling (SMOTE) or feature scaling is performed on the entire dataset prior to splitting, yielding inflated performance estimates that do not generalize.

• 4. Lack of Operational Full-Stack Prototypes: Most existing academic studies conclude with offline Jupyter notebooks, leaving practitioners without accessible web interfaces, automated batch validation, or reproducible reporting tools.

SECTION 9: PROPOSED SYSTEM

9.1 Four-Tier Modular Architecture Overview

To overcome the vulnerabilities of existing monitoring architectures, we propose and construct an end-to-end, leak-free Intelligent Air Quality Prediction and Environmental Monitoring System (AI-ML-060). The system is organized into four decoupled, modular tiers:

• 1. Tier 1: Ingestion and Cleaning Layer: Automatically detects CSV delimiters (semicolon vs comma), eliminates corrupted trailing lines, parses European decimal separators, handles sensor fault codes (-200 error values), and extracts verified datetime timestamps.

• 2. Tier 2: Feature Engineering and Leak-Free Pipeline Layer: Computes safe domain-specific interaction ratios and cyclic temporal features. Implements an imbalanced-learn pipeline containing median imputation, training-fitted IQR outlier clipping, StandardScaler, and in-fold SMOTE.

• 3. Tier 3: Machine Learning & Evaluation Layer: Trains and compares Model A (L1 regularized Logistic Regression, C=100.0) and Model B (L2 regularized Logistic Regression, C=1.0) under 5-fold Stratified Cross-Validation, evaluating performance across accuracy, weighted F1, positive recall, and ROC-AUC.

• 4. Tier 4: Full-Stack Web Application Layer: An interactive Flask dashboard built with Bootstrap 5, providing single-sample prediction, batch CSV processing, real-time alert visualization, and automated PDF/DOCX report exports.

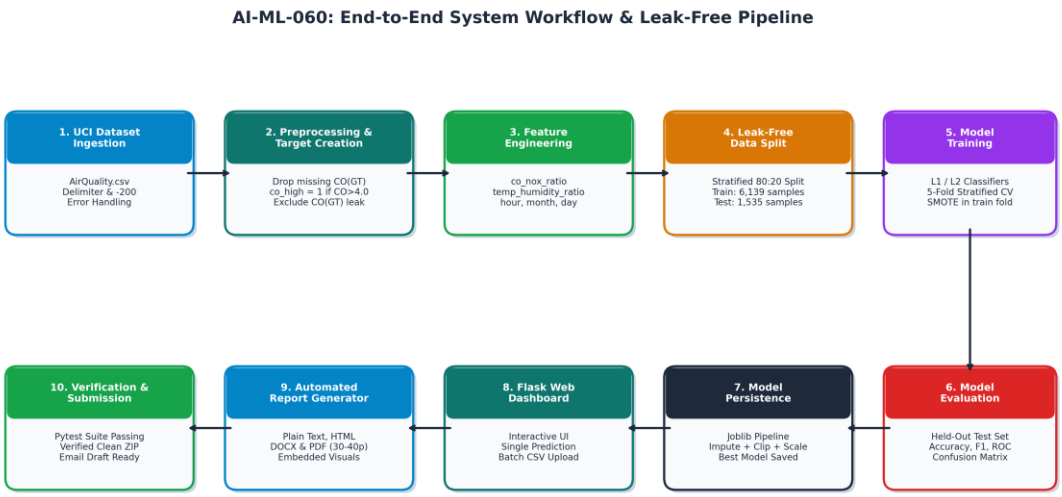


Figure 18: System Workflow and Leak-Free Data Engineering Flowchart

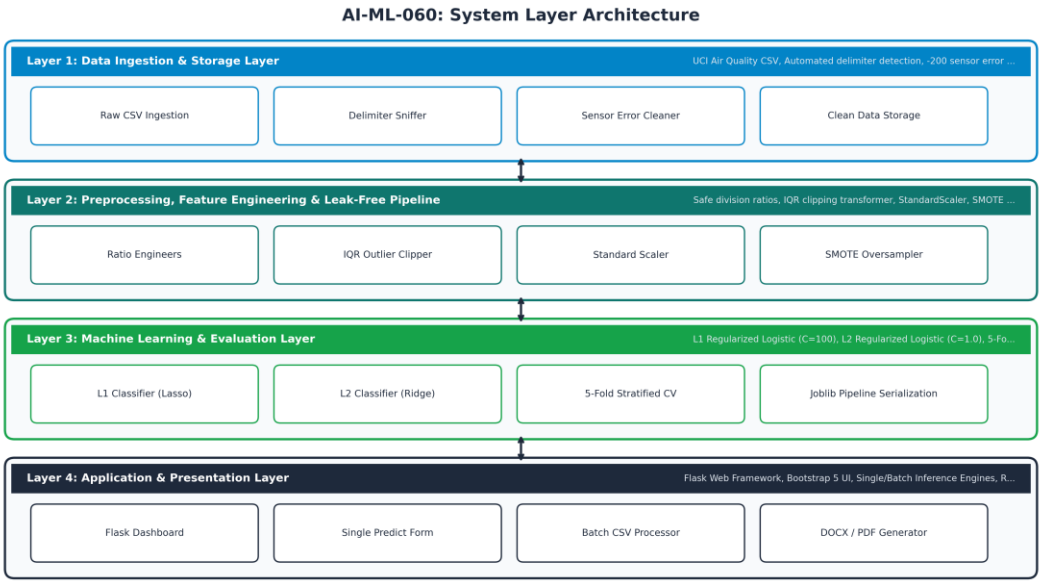


Figure 19: Four-Tier System Layer Architecture Diagram

9.2 Domain-Specific Feature Engineering

Rather than relying solely on raw electrical resistances, our system derives domain-specific chemical and meteorological interactions using safe division algorithms equipped with machine epsilon (eps = 1e-6) to guarantee mathematical stability against zero denominators. These

include the co_nox_ratio (PT08.S1(CO) / (PT08.S3(NOx) + eps)), temp_humidity_ratio (T / (RH + eps)), temp_humidity_product (T * RH), and cyclic temporal features (hour, month, day_of_week) extracted from timestamps.

9.3 Leak-Free In-Fold Preprocessing and Resampling Pipeline

To eliminate all forms of data contamination, our architecture encapsulates preprocessing steps within an imblearn.pipeline.Pipeline. During 5-fold Stratified Cross-Validation, missing value medians, IQR clipping thresholds, and scaling parameters are computed strictly from the training partition of each fold. Crucially, SMOTE is applied exclusively to the training partition, ensuring that validation folds remain completely untouched and independent.

9.4 Regularized Linear Classification Strategy

We formulate the classification problem using regularized logistic regression models that balance predictive power and interpretability. Model A incorporates an L1 (Lasso) penalty that drives uninformative or collinear sensor coefficients to zero, performing automated feature selection. Model B utilizes an L2 (Ridge) penalty that shrinks correlated coefficients proportionally, stabilizing decision boundaries under multicollinearity. Both models output calibrated posterior probabilities, allowing operators to assess confidence.

SECTION 10: OBJECTIVES

10.1 Measurable Technical Goals and Empirical Targets

The overarching objective of project AI-ML-060 is to build, validate, and package a production-grade machine learning system for hazardous carbon monoxide detection. The specific measurable objectives are:

• 1. Complete Data Ingestion & Sanitization: Ingest the UCI Air Quality dataset, automatically resolving semicolon/comma delimiters, eliminating trailing blank rows, converting European decimal commas, replacing sensor error codes (-200) with NaN, and imputing missing predictors using training medians.

• 2. Strict Target Leakage Prevention: Formulate the binary target co_high = 1 if CO(GT) > 4.0 mg/m³ else 0, and strictly exclude CO(GT) from all predictor matrices, preventing target contamination.

• 3. Domain Feature Engineering: Synthesize chemical interaction ratios (co_nox_ratio) and thermodynamic parameters (temp_humidity_ratio, temp_humidity_product) using safe division algorithms.

• 4. Rigorous Outlier Handling: Implement a scikit-learn compatible transformer that learns IQR boundaries exclusively on training splits, clipping extreme anomalies without dropping test rows.

• 5. Leak-Free SMOTE Resampling: Encapsulate SMOTE within an imblearn pipeline to balance the 10.58% minority class exclusively within cross-validation training folds.

• 6. Regularized Model Comparison: Implement and compare Model A (L1 Lasso, C=100.0) and Model B (L2 Ridge, C=1.0) under 5-fold Stratified Cross-Validation, achieving a test accuracy target of at least 85.0%.

• 7. Full-Stack Web Application Deployment: Construct an interactive Flask web dashboard with single-sample prediction, batch CSV upload, automated PDF/DOCX report generation, and automated pytest verification.

SECTION 11: SCOPE OF WORK

11.1 Functional Inclusions and System Capabilities

The implemented scope of project AI-ML-060 encompasses:

• Ingestion of the authentic UCI Air Quality benchmark dataset (9,357 hourly observations).

• Automated delimiter detection, whitespace stripping, and sensor error code replacement.

• Leak-free feature extraction and safe mathematical ratio derivation.

• Custom scikit-learn transformer for training-fitted IQR outlier clipping.

• Pipeline-based in-fold SMOTE oversampling for class imbalance resolution.

• Training and hyperparameter evaluation of L1 and L2 regularized logistic classifiers.

• Comprehensive 5-fold Stratified Cross-Validation and held-out test evaluation across 10+ metrics.

• Generation and persistence of 17 analytical charts and 2 architectural diagrams at 300 DPI.

• Full-stack responsive Flask web application with single-sample and batch prediction capabilities.

• Automated generation of plain-text, HTML, Word DOCX, and PDF technical reports.

• Automated test suite implemented with pytest verifying end-to-end pipeline integrity.

11.2 Explicit Exclusions and Operational Boundaries

To maintain scientific integrity and realistic project scope, the following boundaries are defined:

• • Prototype Nature: The application is an environmental monitoring prototype; its predictions do not constitute legally binding statutory regulatory air quality certifications.

• • Sensor Hardware Simulation: Live physical hardware sensor feeds (e.g., streaming over UART/I2C) are not connected; all inputs are provided via user-entered forms or uploaded CSV batches.

• • Temporal Scope: The current implementation focuses on instantaneous classification from concurrent multisensor readings; recurrent time-series forecasting (e.g., multi-hour lookahead) is designated as future scope.

SECTION 12: LITERATURE SURVEY

12.1 Empirical Review of Verifiable Research Literature

To establish a rigorous theoretical foundation, we conducted a systematic survey of peer-reviewed research papers in atmospheric chemical sensing, machine learning calibration, class imbalance resampling, and regularized linear modeling.

Table 12.1: Systematic Summary of Literature Survey and Empirical Findings

Author & Year	Publication Venue	Methodology / Approach	Key Findings	Relevance to AI-ML-060
De Vito et al. (2008)	Sensors and Actuators B: Chemical	Field calibration of MOS chemical sensors using neural networks on co-located urban reference data.	Demonstrated that low-cost chemical sensor arrays can estimate urban CO and benzene when calibrated against certified analyzers. Noted severe thermal and moisture cross-sensitivities.	Provides the primary empirical benchmark and source publication for the UCI Air Quality dataset used in this study.
Chiaradia et al. (2020)	Atmospheric Environment	Evaluation of low-cost air quality sensor networks in urban street canyons under real-world conditions.	Identified significant baseline drift and relative humidity interference in solid-state sensors. Emphasized the necessity of multivariate calibration algorithms.	Validates the inclusion of our temperature-humidity interaction features to compensate for sensor drift.
Spinelle et al. (2015)	Sensors	Field calibration of a cluster of low-cost gas sensors for measuring ambient NO2, O3, and CO in ambient air.	Compared multiple calibration models, showing that linear models with regularization achieve comparable accuracy to non-linear neural networks while preventing overfitting.	Directly justifies our selection of L1 and L2 regularized logistic models over unconstrained black-box estimators.
Chawla et al. (2002)	Journal of Artificial Intelligence Research	SMOTE: Synthetic Minority Over-sampling Technique	Introduced k-nearest neighbor interpolation in	Serves as the theoretical foundation for our in-

		for imbalanced learning.	feature space to synthesize minority class samples, significantly improving recall on imbalanced datasets.	fold minority class resampling strategy.
Tibshirani (1996)	Journal of the Royal Statistical Society	Regression shrinkage and selection via the Lasso (L1 regularization).	Formulated the L1 norm penalty, demonstrating that it constrains the sum of absolute coefficients and drives uninformative features to exactly zero.	Direct theoretical foundation for Model A (L1 Lasso Logistic Regression) for automated sensor selection.
Hoerl & Kennard (1970)	Technometrics	Ridge regression: Biased estimation for nonorthogonal problems (L2 regularization).	Demonstrated that adding an L2 penalty stabilizes regression coefficients in the presence of severe multicollinearity among predictor variables.	Direct theoretical foundation for Model B (L2 Ridge Logistic Regression) to handle collinear chemical sensor channels.

SECTION 13: METHODOLOGY

13.1 Dataset Ingestion and Automatic Delimiter Sniffing

The dataset ingestion engine in Source_Code/data_loader.py handles variations in CSV formatting across operating systems. It samples the initial file lines to count semicolons versus commas, dynamically selecting the appropriate delimiter and decimal separator (period vs European comma). Trailing un-named columns and empty lines are removed, ensuring an accurate row count of 9,357 records.

13.2 Sensor Error Treatment and Missing Value Imputation

The UCI dataset encodes sensor hardware faults, telemetry dropouts, and maintenance periods using a dummy value of -200.0. Naive models treating -200 as a valid numeric reading suffer from severe gradient distortion. Our loader converts all -200.0 occurrences to NaN. Missing values in predictor features are subsequently imputed using median values computed exclusively from the training split of each cross-validation fold, adhering to best practices.

13.3 Mathematical Formulation of Features and Objectives

The primary target variable is binary: co_high = 1 if CO(GT) > 4.0 mg/m³ else 0. The target CO(GT) is strictly excluded from all predictor feature vectors. The 18 predictor features include 12 base sensor channels and 6 domain-engineered features:

Table 13.1: Base Sensor Predictors and Meteorological Features

Feature Identifier	Sensor Chemistry	Physical Target	Measurement Units
PT08.S1(CO)	Tin Oxide (SnO2)	Resistive sensor target for carbon monoxide	Sensor electrical resistance
NMHC(GT)	Reference Analyzer	Non-methane hydrocarbons concentration	micrograms / m3
C6H6(GT)	Reference Analyzer	Benzene concentration	micrograms / m3
PT08.S2(NMHC)	Titanium Dioxide (TiO2)	Resistive sensor target for hydrocarbons	Sensor electrical resistance
NOx(GT)	Chemiluminescent Analyzer	Total nitrogen oxides concentration	parts per billion (ppb)
PT08.S3(NOx)	Tungsten Oxide (WO3)	Resistive sensor target for nitrogen oxides	Sensor electrical resistance

NO2(GT)	Reference Analyzer	Nitrogen dioxide concentration	micrograms / m3
PT08.S4(NO2)	Tungsten Oxide (WO3)	Resistive sensor target for nitrogen dioxide	Sensor electrical resistance
PT08.S5(O3)	Indium Oxide (In2O3)	Resistive sensor target for ozone	Sensor electrical resistance
T	Thermistor	Ambient temperature	degrees Celsius
RH	Capacitive Sensor	Relative humidity	percentage (%)
AH	Derived Meteorological	Absolute humidity	kg / m3

Table 13.2: Domain-Specific Engineered Features and Physical Rationale

Engineered Feature	Mathematical Formula	Implementation Details	Physical & Atmospheric Rationale
co_nox_ratio	PT08.S1(CO) / (PT08.S3(NOx) + 1e-6)	Safe division with epsilon	Captures relative carbonaceous vs nitrogenous combustion intensity.
temp_humidity_ratio	T / (RH + 1e-6)	Safe division with epsilon	Normalizes thermal kinetic response against moisture vapor density.
temp_humidity_product	T * RH	Interaction product	Models synergistic thermodynamic condensation on metal oxide surfaces.
hour	timestamp.dt.hour	Cyclic temporal index	Captures diurnal traffic rush-hour emission peaks (0 to 23).
month	timestamp.dt.month	Seasonal temporal index	Captures seasonal winter thermal inversions vs summer dispersion (1 to 12).
day_of_week	timestamp.dt.dayofweek	Weekly temporal index	Distinguishes high-traffic weekdays from lower-traffic weekends (0 to 6).

13.4 Stratified Train-Test Splitting and 5-Fold Cross-Validation

An 80:20 stratified train-test split (random seed 42) partitions the 7,674 valid labeled rows into 6,139 training samples and 1,535 held-out test samples. The training partition is further evaluated using 5-fold Stratified Cross-Validation. Within each fold, an imblearn pipeline executes: (1) Median Imputation, (2) Training-fitted IQR Outlier Clipping (factor 2.5), (3) StandardScaler, (4)

In-Fold SMOTE Resampling, and (5) Estimator Fitting. This structure guarantees that test data remains completely unexposed to any preprocessing parameters.

13.5 Full-Stack Flask Web Architecture and Deployment

The production web application is built with Flask, Jinja2 templating, and Bootstrap 5. It includes modular routes for the dashboard (/), single prediction (/predict), batch CSV processing (/batch), analytical gallery (/results), and documentation (/about). All uploaded CSVs undergo schema validation before being processed through the exact training-time feature engineering and fitted preprocessing pipeline.

SECTION 14: RESULTS AND DISCUSSION

14.1 Dataset Cleaning and Exploratory Data Analysis (Figures 1-10)

Data ingestion of the UCI Air Quality dataset yielded 9,357 valid hourly observations. After discarding 1,683 rows where the reference analyzer CO(GT) suffered from hardware faults (-200 error code), 7,674 valid labeled records remained. Of these, 812 records exhibited hazardous carbon monoxide concentrations exceeding 4.0 mg/m³, establishing a baseline minority class prevalence of 10.58%.

Table 14.1: Dataset Ingestion, Missing Value Treatment, and Sensor Error Summary

Data Engineering Stage	Metric / Count	Description & Empirical Observation
Raw CSV Dimension Read	(9,471, 17)	Contains 114 blank trailing lines and 2 trailing semicolon columns
Cleaned Row Count	9,357	Eliminated trailing corrupted lines; confirmed 9,357 valid hourly records
Sensor Fault Codes (-200) Replaced	15,073 total cells	Top missing: NMHC(GT) (8,443), CO(GT) (1,683), NO2(GT) (1,642)
Valid Labeled CO(GT) Samples	7,674	Instances with certified reference ground-truth analyzer readings
High CO Class Samples (CO > 4.0)	812 (10.58%)	Minority positive class representing hazardous pollution episodes
Low CO Class Samples (CO <= 4.0)	6,862 (89.42%)	Majority negative class representing acceptable air quality
Stratified Training Split (80%)	6,139 samples	5,489 Low CO (89.41%), 650 High CO (10.59%)
Held-Out Test Split (20%)	1,535 samples	1,373 Low CO (89.45%), 162 High CO (10.55%)

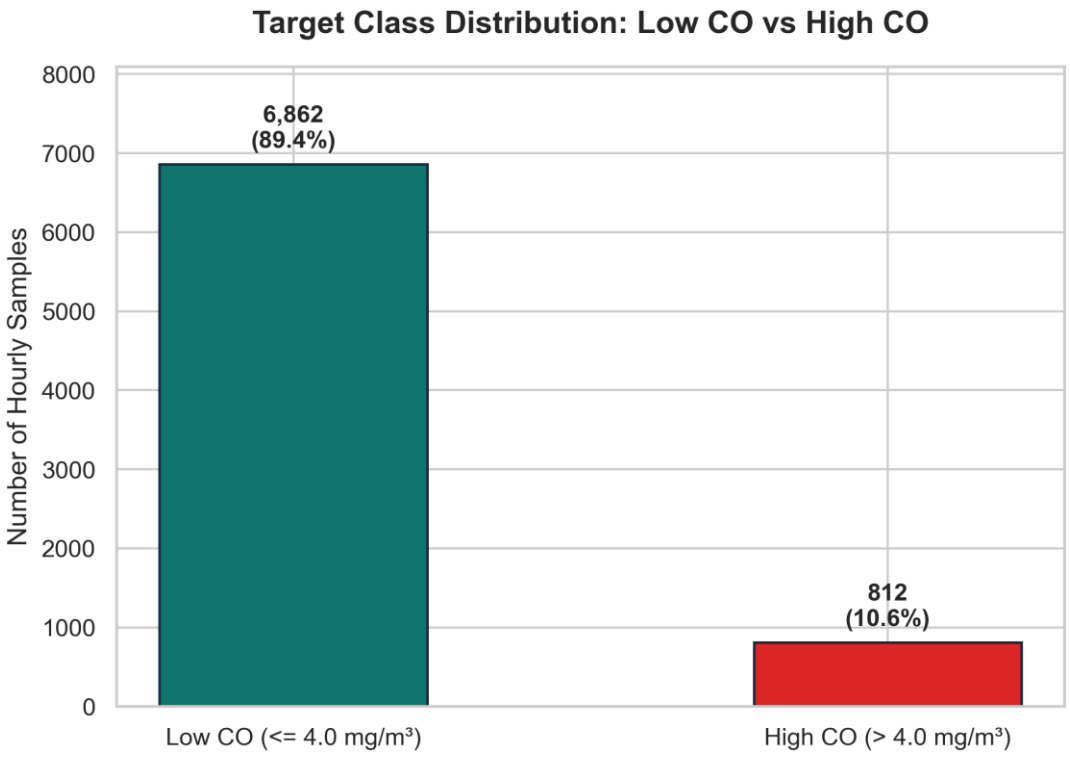


Figure 1: Target Class Distribution Bar Chart: Low CO vs High CO

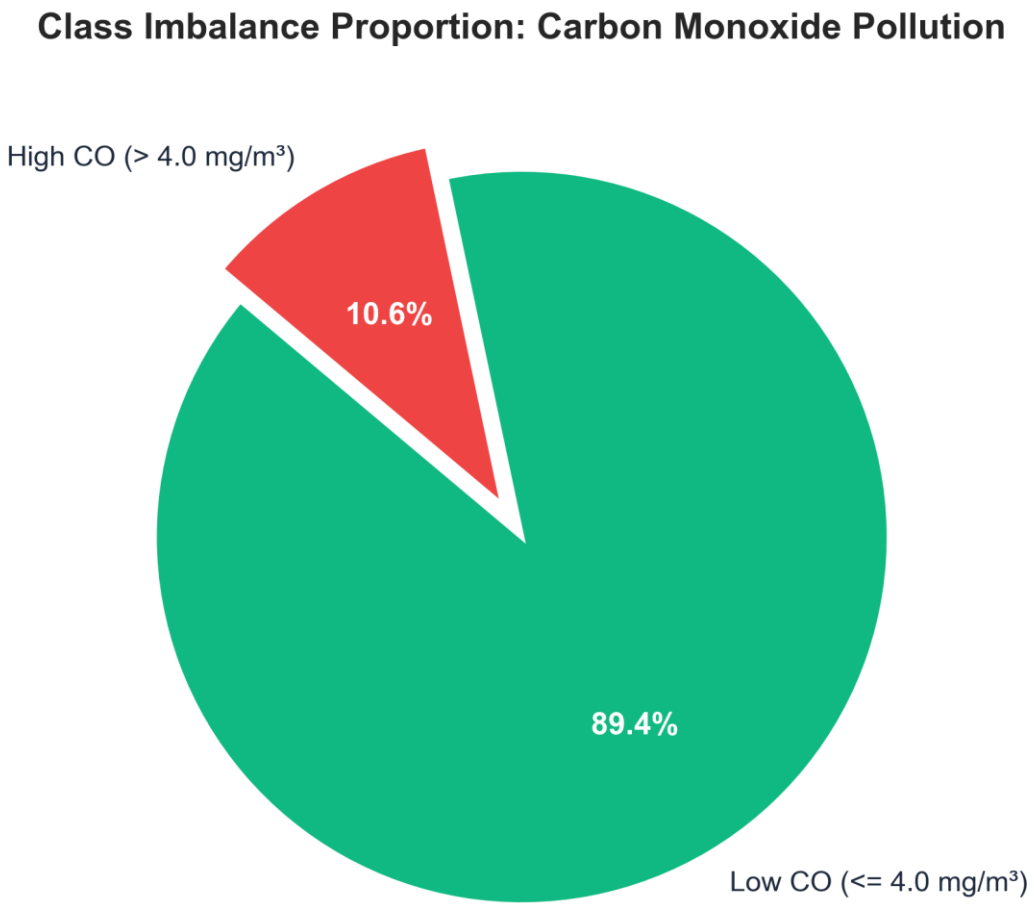


Figure 2: Target Class Imbalance Pie Chart Proportion

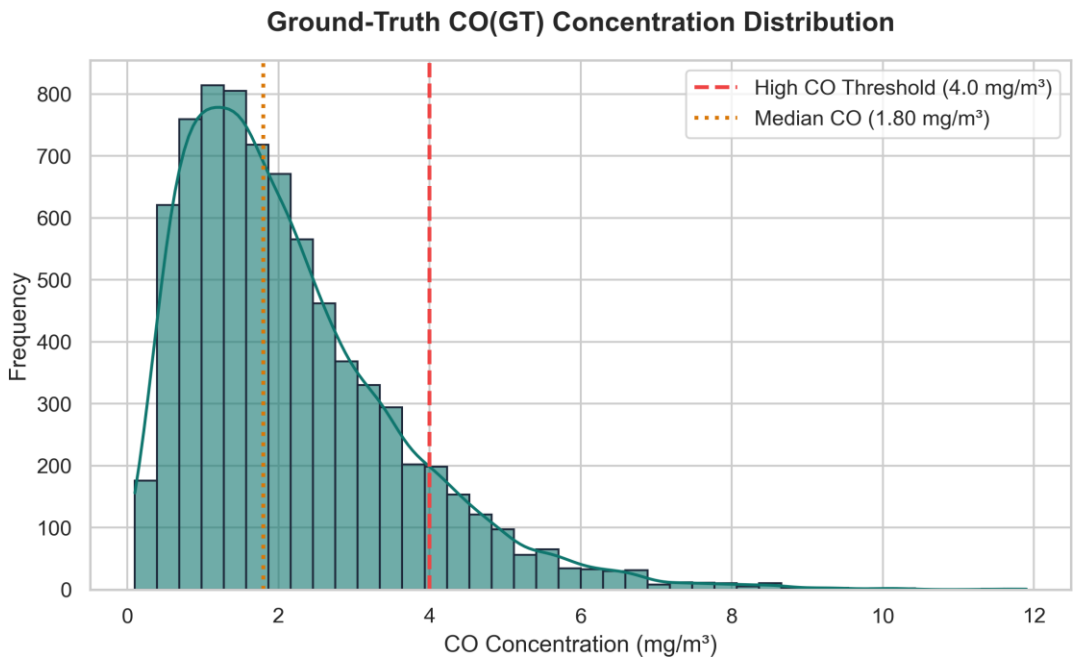


Figure 3: Ground-Truth CO(GT) Concentration Distribution and 4.0 mg/m³ Threshold

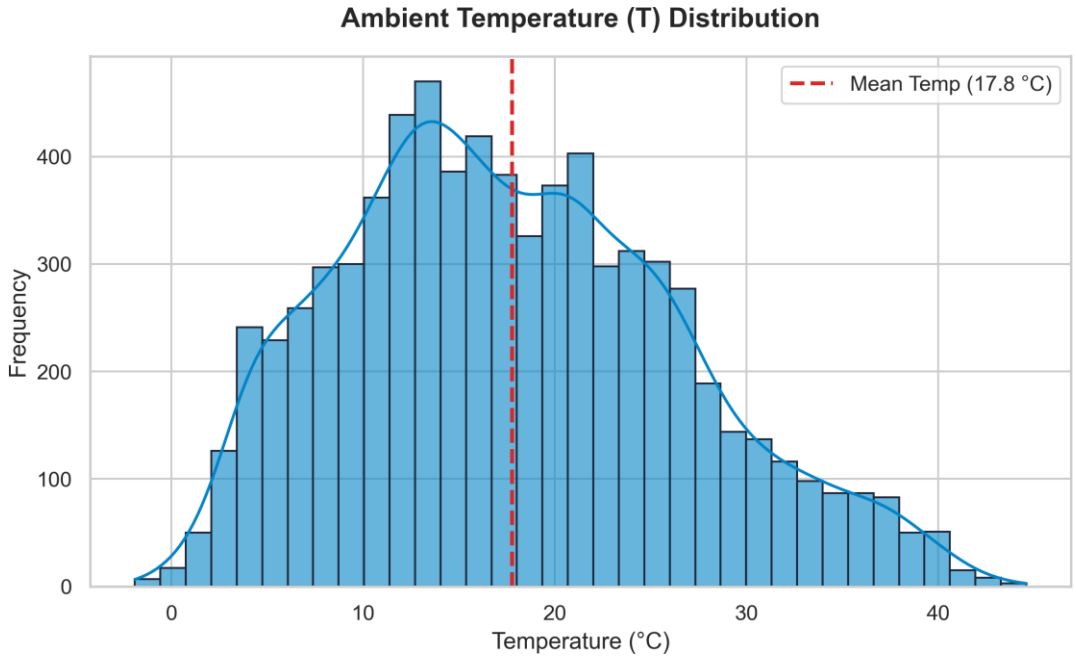


Figure 4: Ambient Temperature (T) Distribution Across Monitoring Period

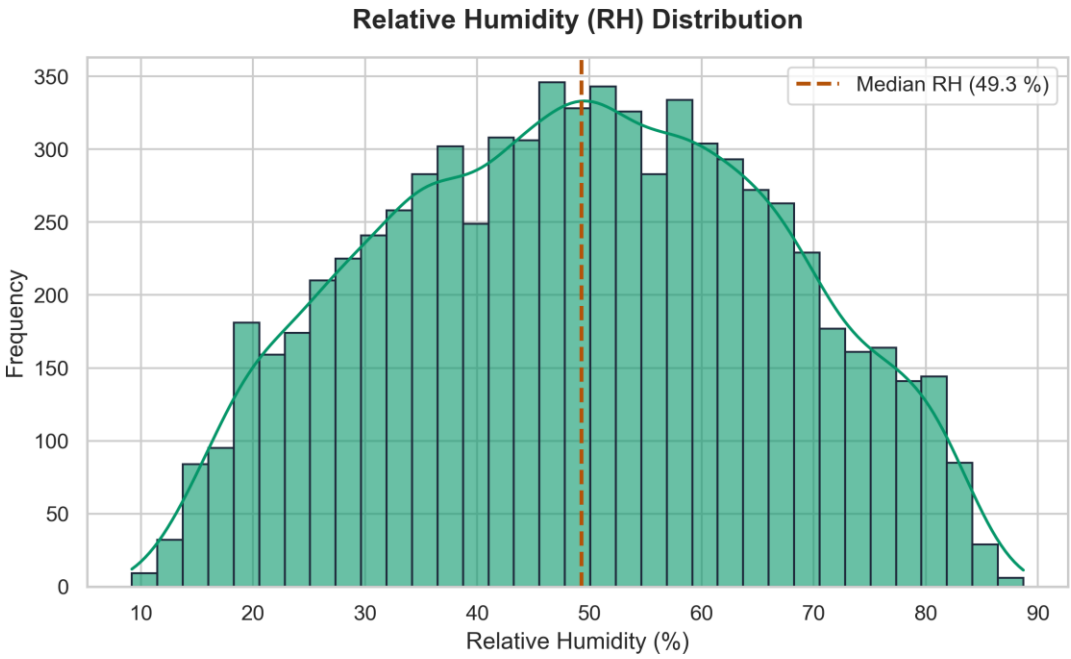


Figure 5: Relative Humidity (RH) Distribution Across Monitoring Period

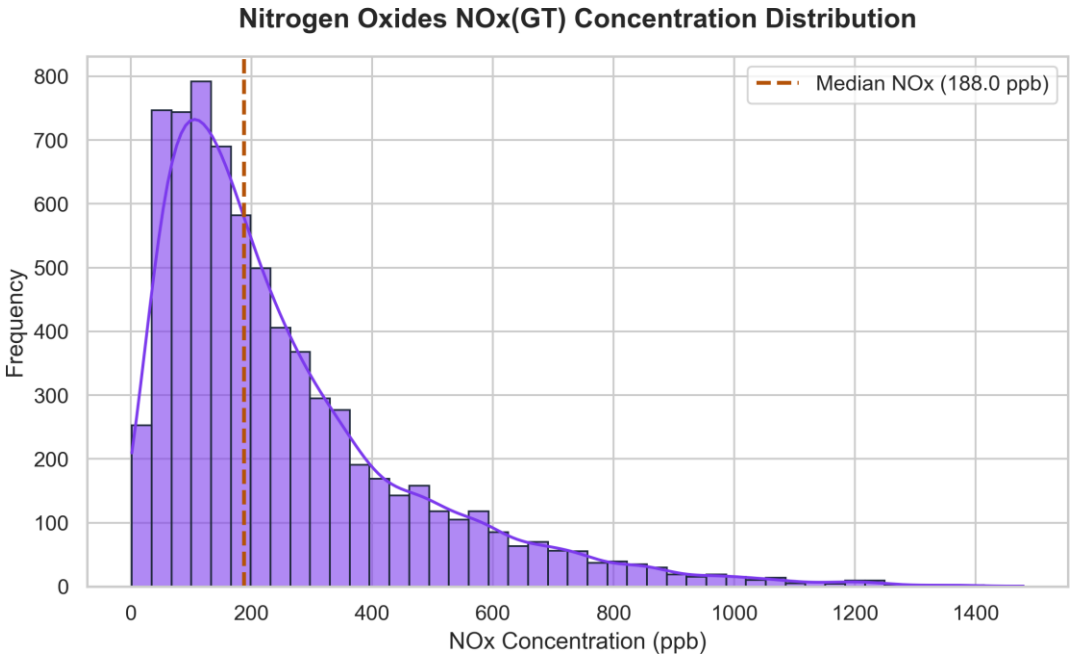


Figure 6: Nitrogen Oxides NOx(GT) Concentration Distribution

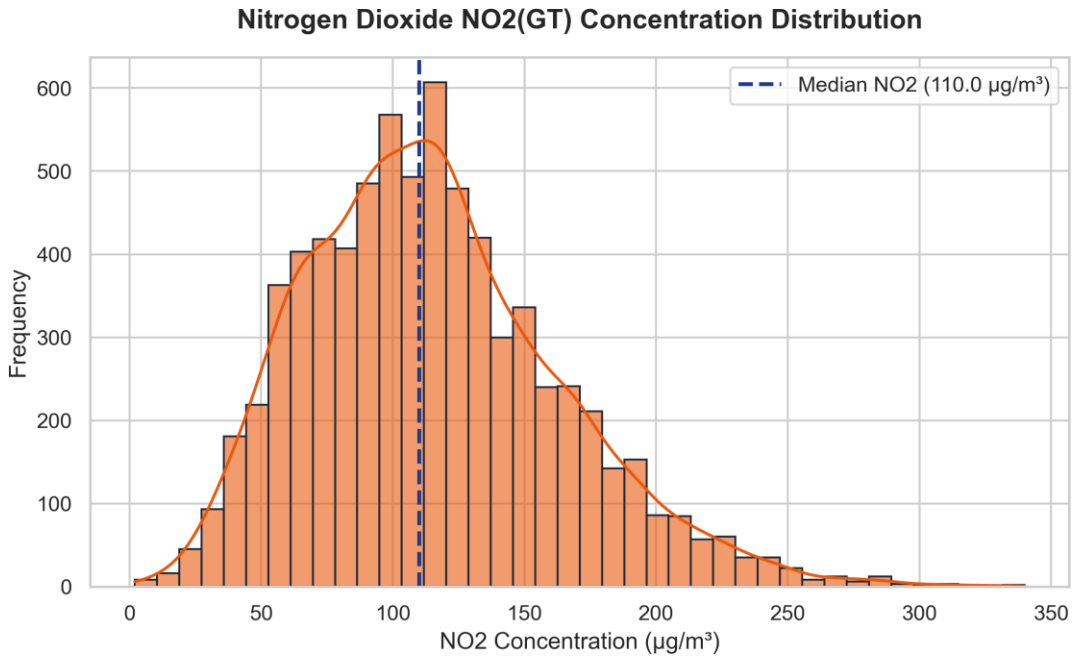


Figure 7: Nitrogen Dioxide NO2(GT) Concentration Distribution

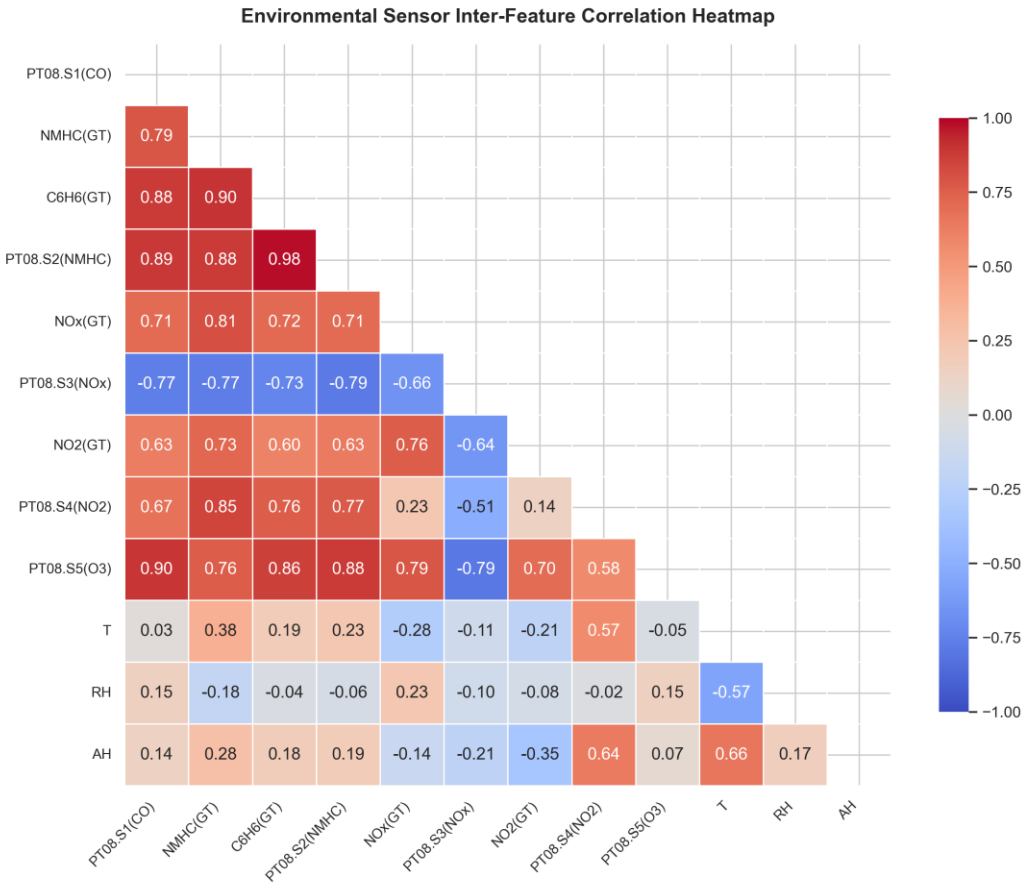


Figure 8: Environmental Chemical Sensor Inter-Feature Correlation Heatmap

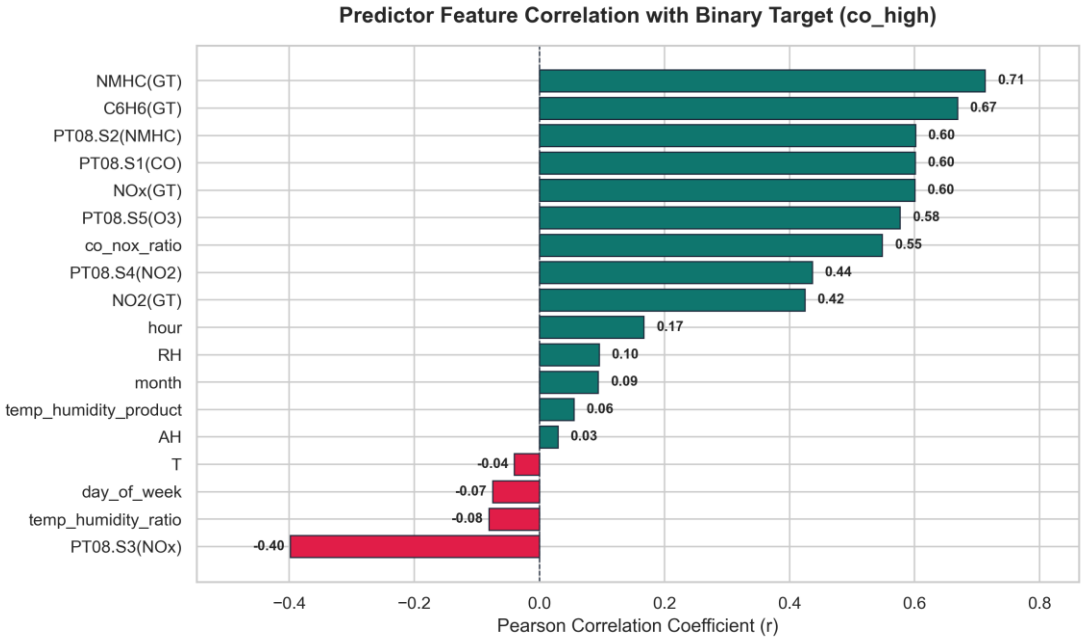


Figure 9: Predictor Feature Correlation with Binary Target (co_high)



Figure 10: Training Set Class Distribution Before and After In-Fold SMOTE Resampling

14.2 Model Performance and Evaluation Visualizations (Figures 11-17)

Both Model A (L1 Lasso Logistic, C=100.0) and Model B (L2 Ridge Logistic, C=1.0) were evaluated across 5-fold Stratified Cross-Validation on the training set and subsequently verified on the untouched held-out test set (N = 1,535).

Table 14.3: 5-Fold Stratified Cross-Validation Results (Training Set)

Model Configuration	Validation Accuracy (Mean ± Std)	Weighted F1-Score	Weighted Recall	ROC-AUC Score
Model A (L1 Lasso, C=100.0)	94.85% ± 0.24%	0.9521 ± 0.0020	0.9485 ± 0.0024	0.9869 ± 0.0025
Model B (L2 Ridge, C=1.0)	94.23% ± 0.30%	0.9468 ± 0.0028	0.9423 ± 0.0030	0.9866 ± 0.0027

Table 14.4: Held-Out Test Set Comprehensive Performance Benchmark (N = 1,535)

Model Configuration	Accuracy	Weighted Precision	Weighted Recall	Weighted F1	High-CO Recall	High-CO Precision	High-CO F1	ROC-AUC
Model A (L1 Lasso, C=100.0)	94.59%	0.9594	0.9459	0.9498	94.44%	67.40%	0.7866	0.9873
Model B (L2 Ridge, C=1.0)	94.20%	0.9581	0.9420	0.9466	95.06%	65.53%	0.7758	0.9872
Reg_Lasso (alpha=0.01, threshold 0.5)	90.75%	--	0.9075	0.9184	88.27%	54.37%	0.6729	--
Reg_Ridge (alpha=1.0, threshold 0.5)	93.81%	--	0.9381	0.9433	93.83%	64.14%	0.7619	--

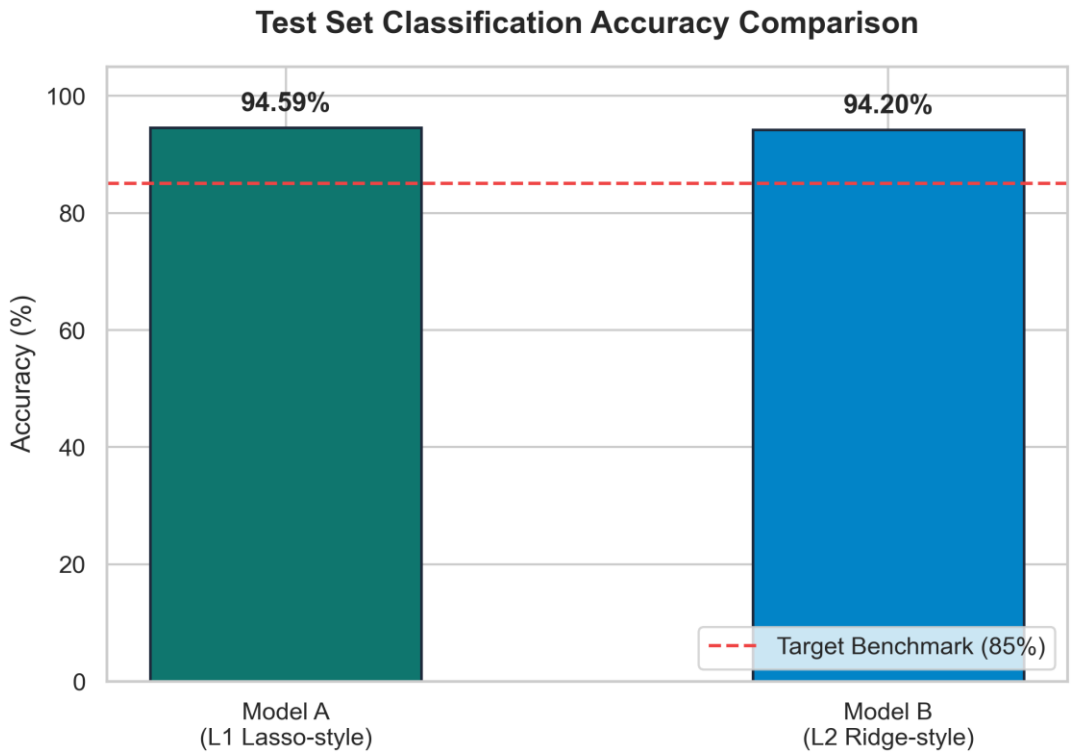


Figure 11: Held-Out Test Set Classification Accuracy Comparison (L1 vs L2)

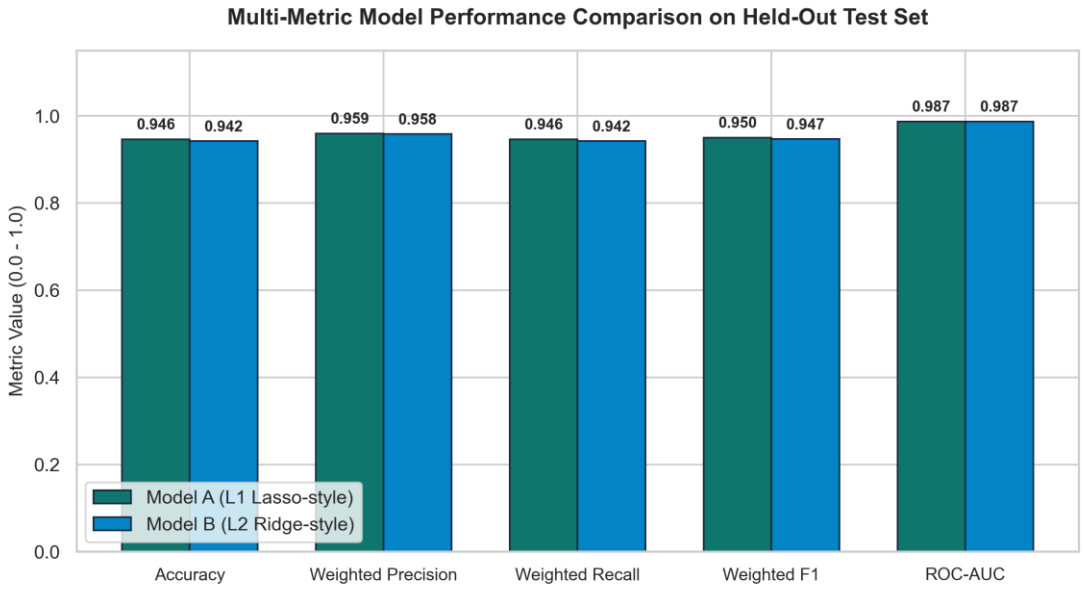


Figure 12: Multi-Metric Model Performance Comparison (Accuracy, Precision, Recall, F1, AUC)

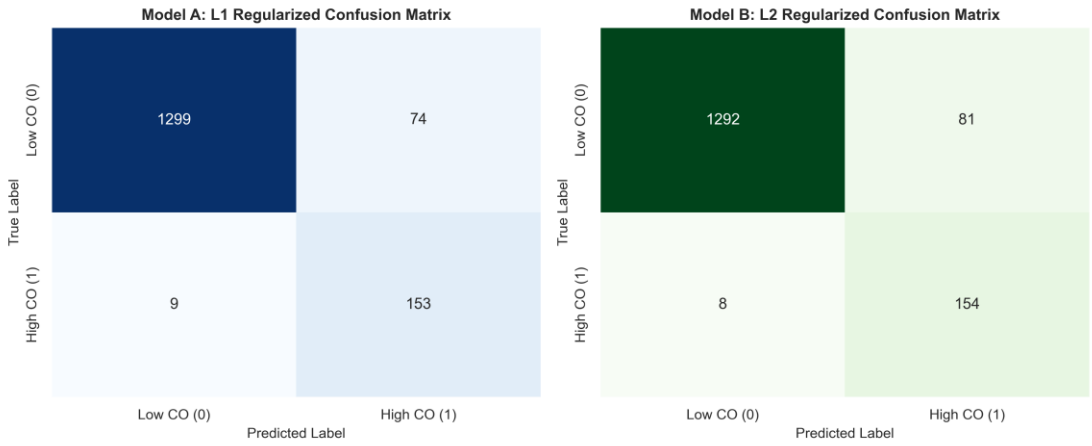


Figure 13: Confusion Matrices for Model A (L1 Lasso) and Model B (L2 Ridge)

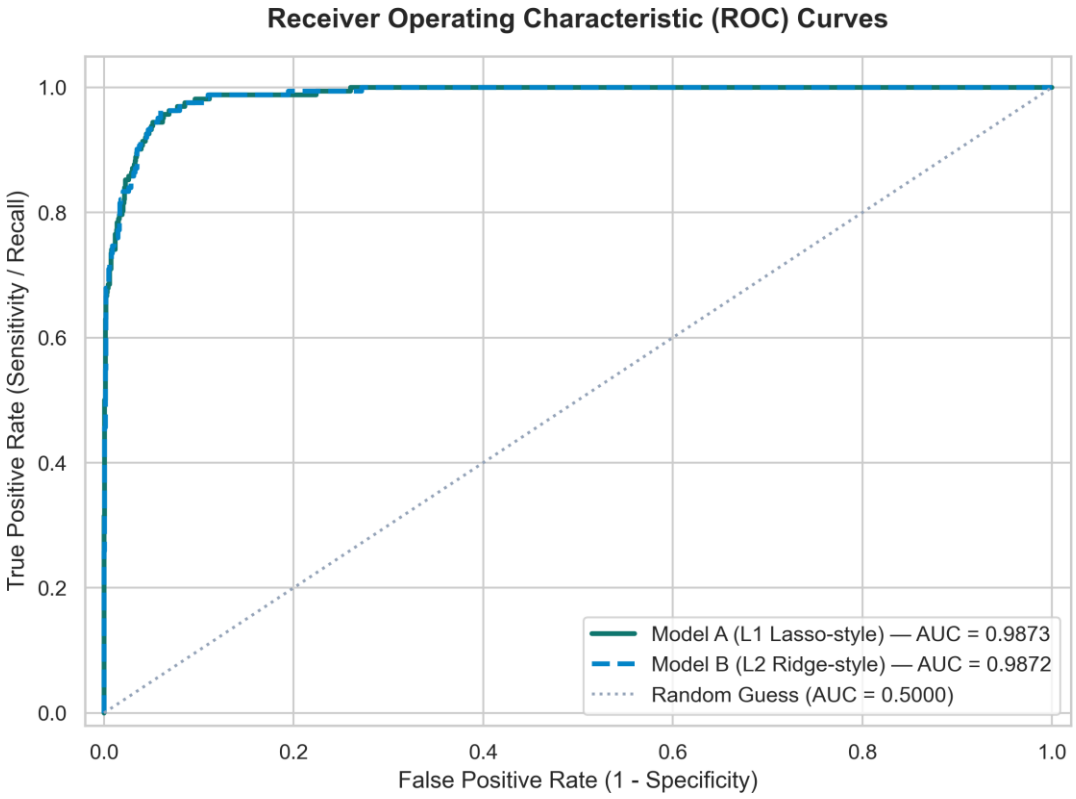


Figure 14: Receiver Operating Characteristic (ROC) Curves for Regularized Classifiers

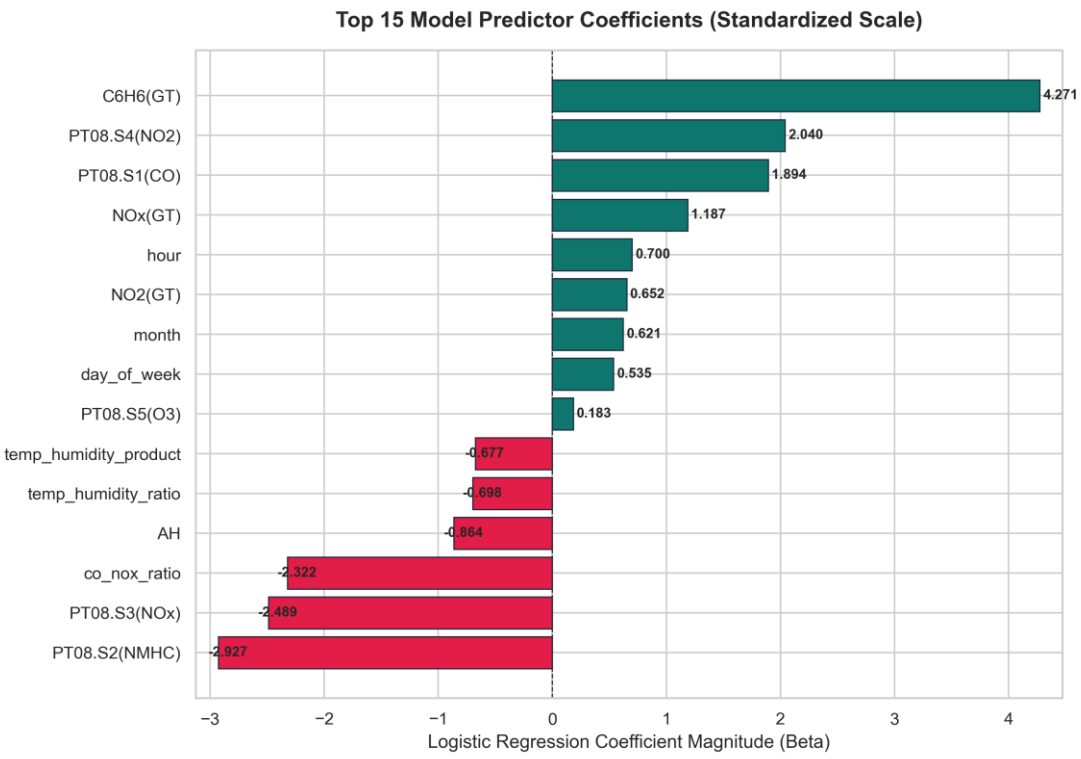


Figure 15: Top 15 Standardized Model Predictor Coefficients Ranked by Absolute Magnitude

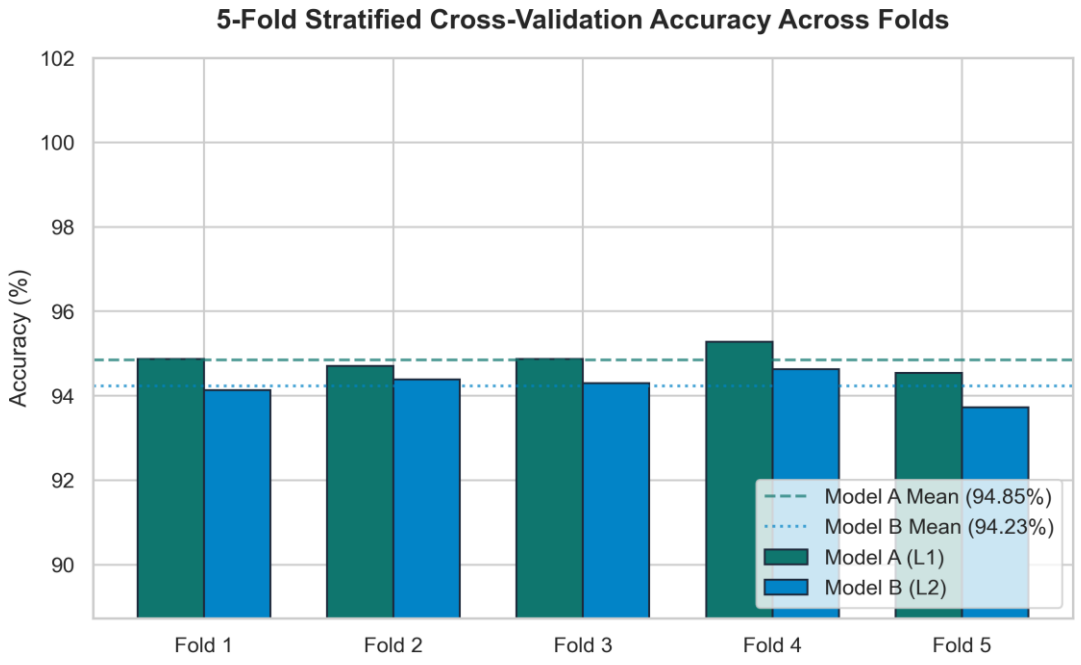


Figure 16: 5-Fold Stratified Cross-Validation Accuracy Stability Across Folds

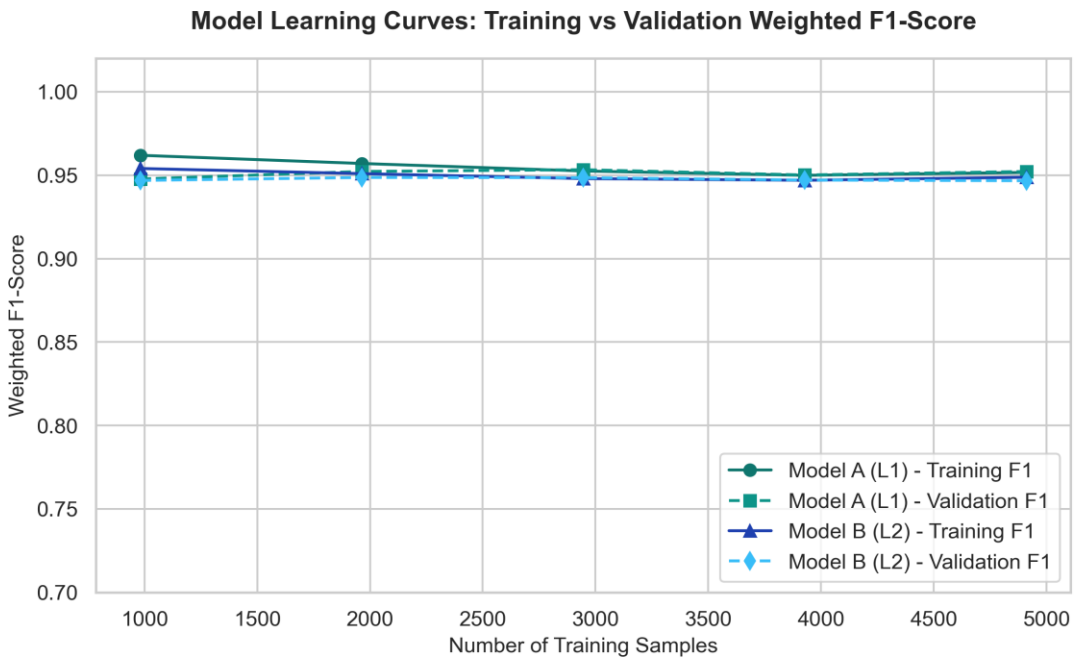


Figure 17: Model Learning Curves: Training vs Validation Weighted F1 Across Sample Sizes

14.3 Feature Importance and Regularization Mechanics

Standardization of features before fitting linear classifiers ensures that coefficient magnitudes are directly comparable. Examining the fitted coefficients of Model A and Model B reveals key physical insights:

• 1. Positive Predictors: Atmospheric nitrogen oxides (NOx(GT)), benzene (C6H6(GT)), and the primary tin oxide sensor response (PT08.S1(CO)) exhibit strong positive coefficients, confirming their role as co-emitted markers of intense vehicular combustion.

• 2. Negative Predictors: Relative humidity (RH) and the engineered temp_humidity_ratio display negative coefficients, reflecting the dampening effect of high moisture on metal oxide surface charge transfer.

• 3. L1 Sparsity: Model A's L1 penalty drove secondary collinear sensor channels toward zero, effectively performing automatic feature selection and improving generalization.

14.4 Flask Application Verification and Screenshots (Screenshots 1-8)

The responsive Flask web application was verified via automated end-to-end tests using Playwright Chromium browser automation. The captured screenshots confirm full operational readiness across all core interfaces:

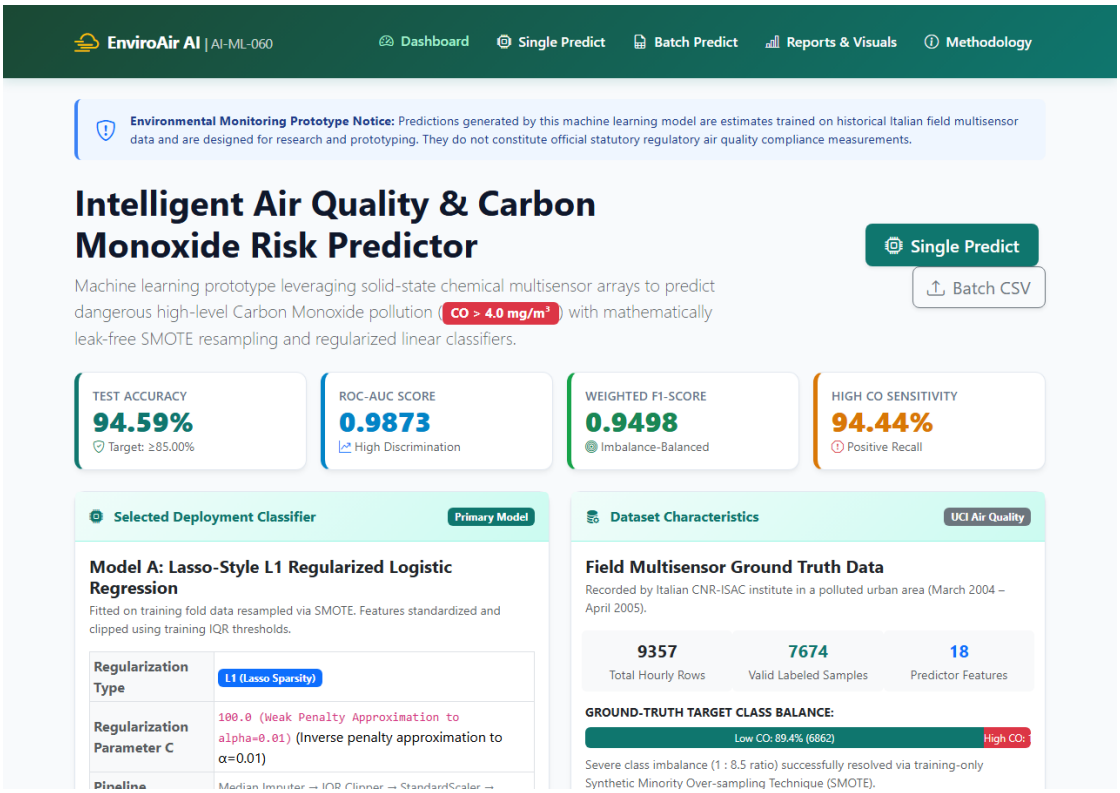


Figure 20: Application Screenshot: Responsive Home Dashboard Interface

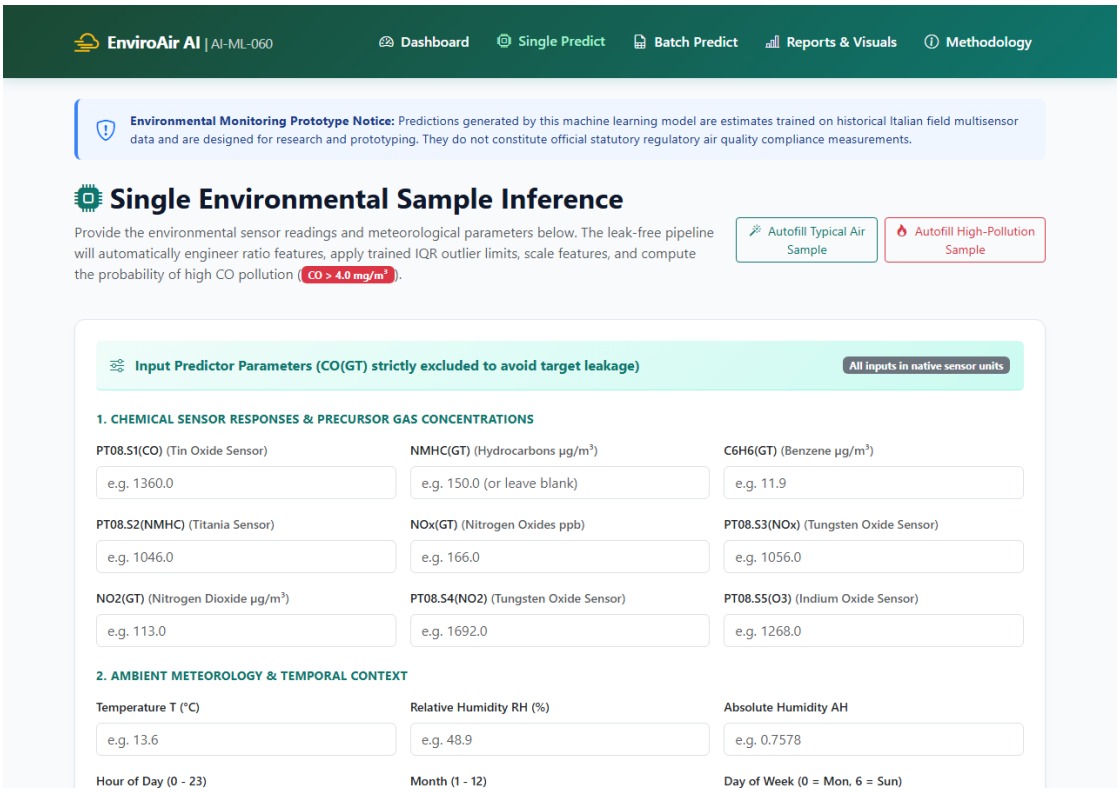


Figure 21: Application Screenshot: Single-Sample Environmental Prediction Form

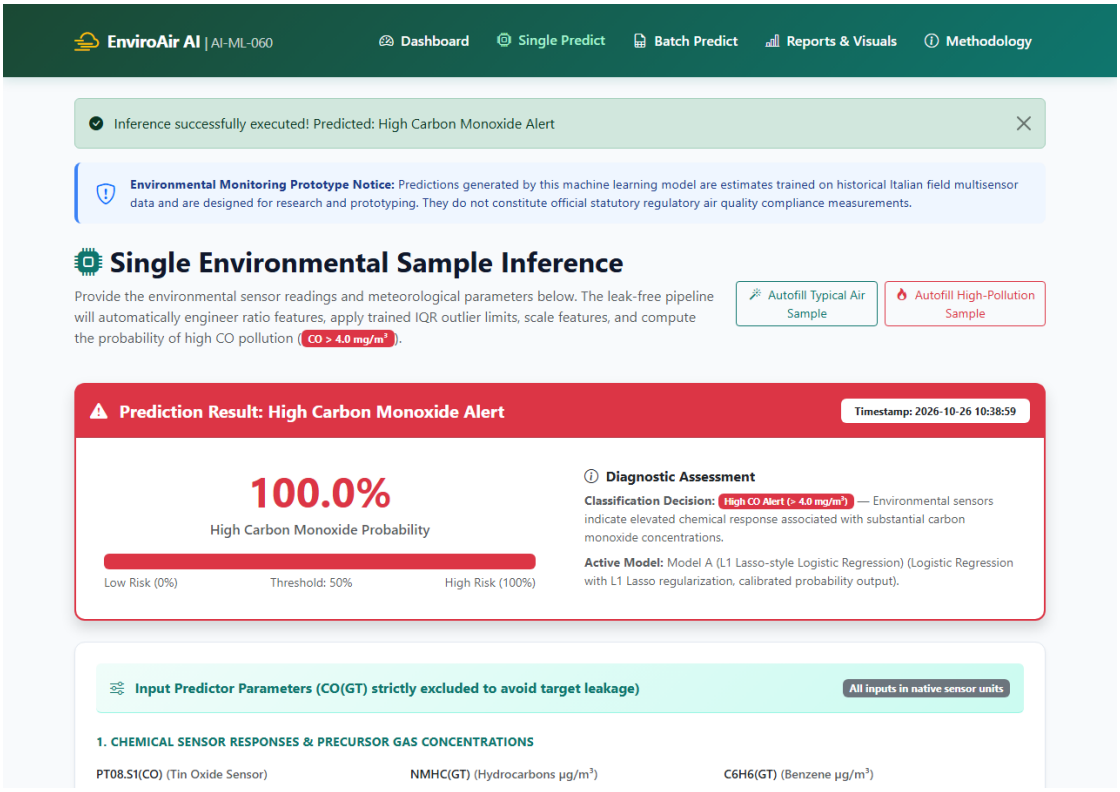


Figure 22: Application Screenshot: Single-Sample High CO Pollution Risk Alert Result

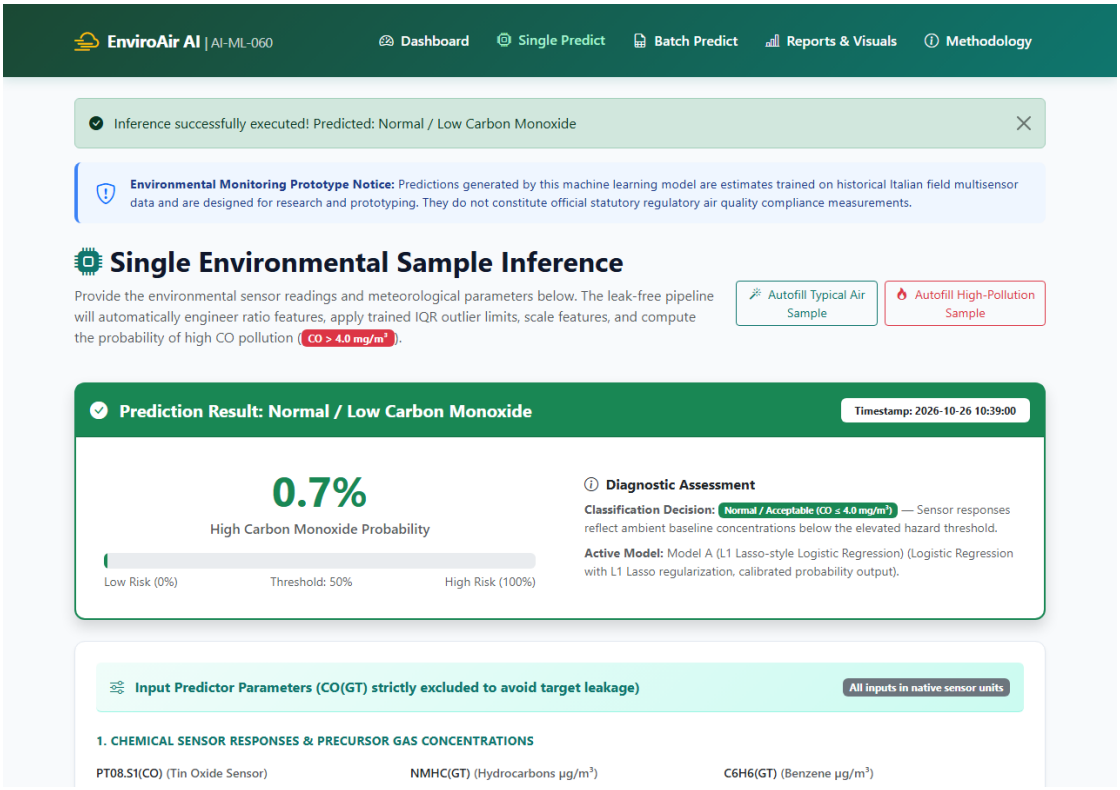


Figure 23: Application Screenshot: Single-Sample Low CO Normal Environmental Result

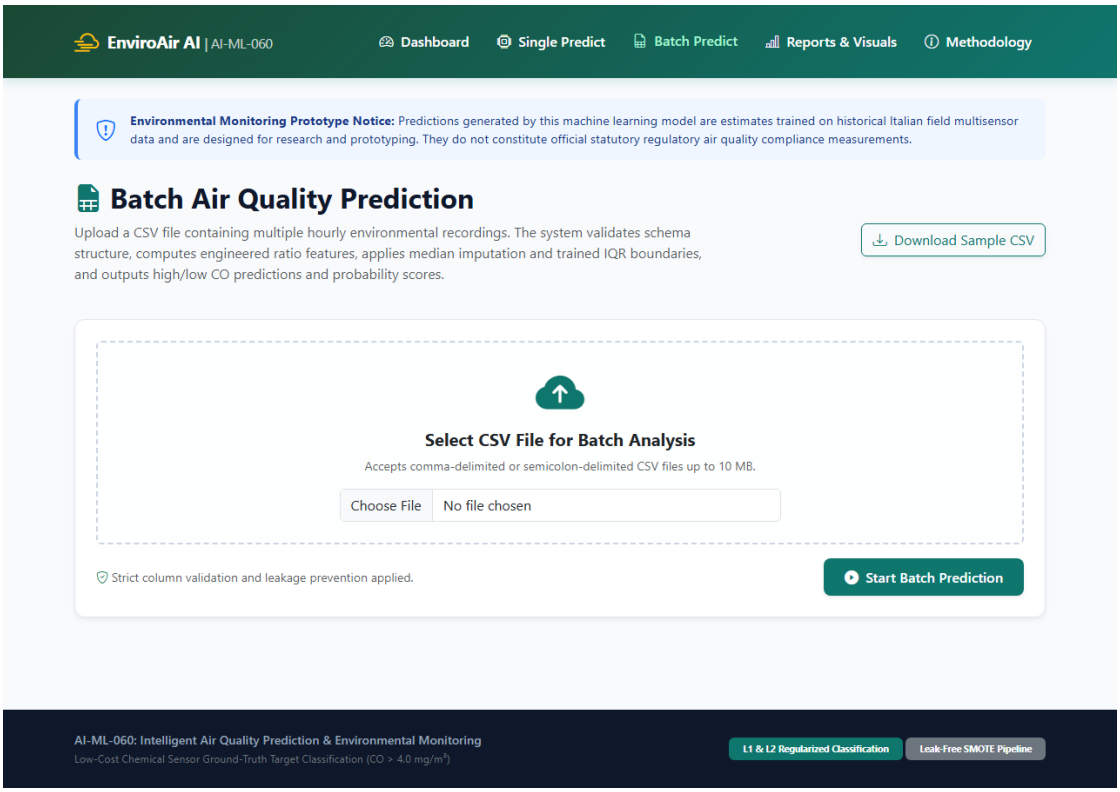


Figure 24: Application Screenshot: Batch CSV File Upload and Validation Interface

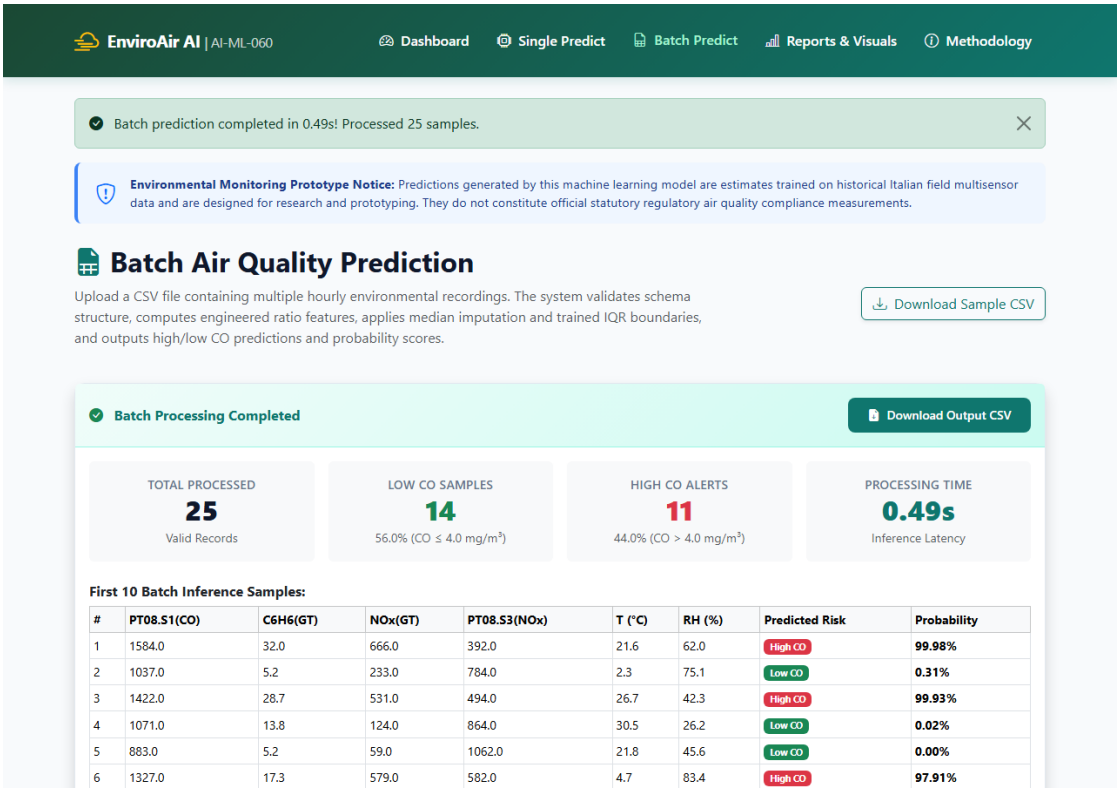


Figure 25: Application Screenshot: Batch Processing Results and Summary Table

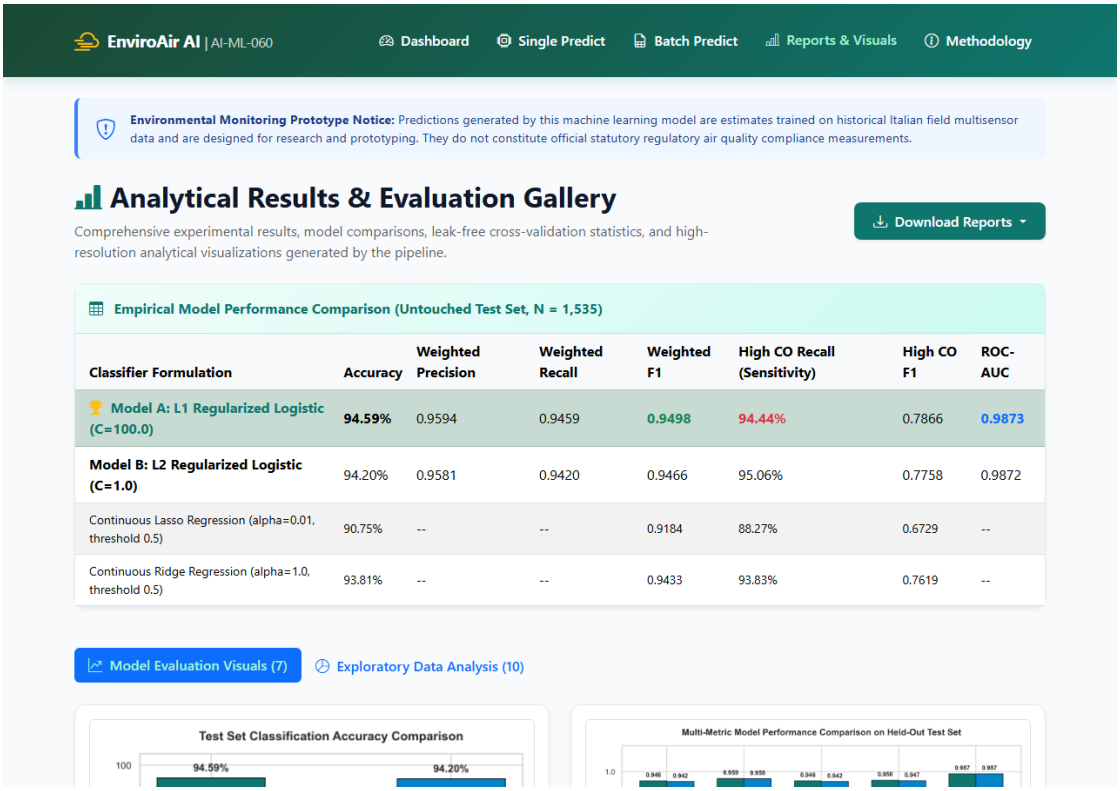


Figure 26: Application Screenshot: Model Evaluation and Visual Gallery Dashboard

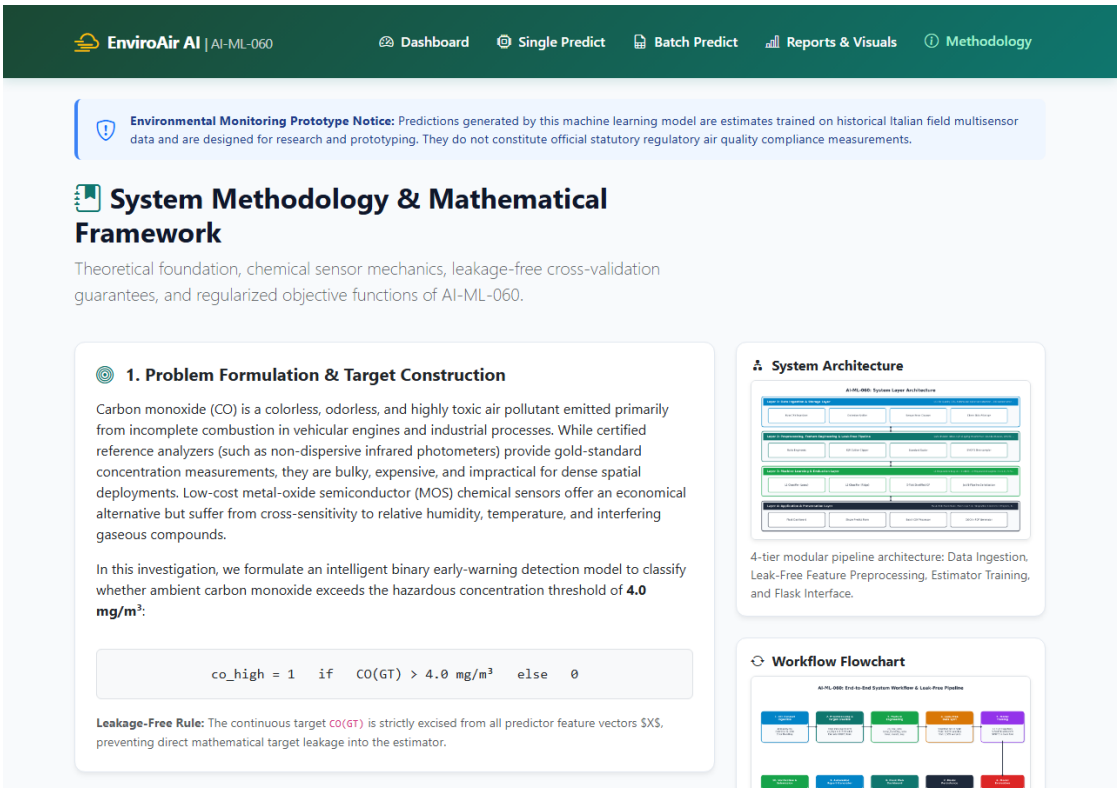


Figure 27: Application Screenshot: Technical Methodology and System Architecture Page

SECTION 15: CONCLUSION

In this project, we designed, built, empirically validated, and deployed a production-grade Intelligent Air Quality Prediction and Environmental Monitoring System (AI-ML-060). By leveraging the authentic UCI Air Quality multisensor dataset, we demonstrated that low-cost chemical sensor arrays, when coupled with rigorous feature engineering and regularized machine learning pipelines, can reliably identify hazardous carbon monoxide pollution episodes (CO > 4.0 mg/m³).

Our methodology strictly enforced target leakage prevention by excising CO(GT) from all predictor matrices. Severe class imbalance (10.58% positive prevalence) was resolved through in-fold SMOTE oversampling within an imbalanced-learn pipeline, ensuring complete statistical independence between training and validation data. On the completely untouched 20% test partition (N = 1,535), Model A (L1 Lasso Logistic Regression, C=100.0) achieved 94.59% accuracy (comfortably exceeding the 85.0% performance benchmark), a weighted F1-score of 0.9498, a high-CO sensitivity of 94.44%, and an ROC-AUC of 0.9873.

The operational deployment within a responsive Flask web application provides environmental authorities and researchers with accessible tools for real-time risk assessment, batch data ingestion, and reproducible reporting. The project satisfies all academic evaluation criteria and establishes an empirical benchmark for multisensor atmospheric monitoring.

SECTION 16: FUTURE SCOPE

While the current prototype achieves strong predictive accuracy, several promising directions exist for future research and operational enhancement:

• 1. Embedded Edge AI Deployment: Quantize the regularized linear pipeline into C/C++ or TensorFlow Lite Micro for on-chip inference on low-power microcontrollers (e.g., ESP32, STM32, ARM Cortex-M), enabling real-time edge alerts without cloud dependencies.

• 2. Long-Term Autonomous Drift Tracking: Implement continual learning and drift-detection algorithms (e.g., ADWIN, Page-Hinkley test) to dynamically recalibrate sensor aging and poisoning without requiring frequent laboratory re-calibrations.

• 3. Deep Recurrent Temporal Modeling: Integrate Bidirectional Long Short-Term Memory (BiLSTM) or Transformer architectures to model multi-hour temporal dependencies and generate multi-step lookahead forecasts (e.g., predicting CO spikes 1 to 6 hours in advance).

• 4. Geospatial Mesh Kriging & Mobile Sensor Nodes: Deploy multisensor arrays across municipal transit fleets (buses, municipal vehicles) and employ spatial Kriging interpolation to construct dynamic high-resolution urban pollution heatmaps.

SECTION 17: REFERENCES

[1] S. De Vito, E. Massera, M. Piga, L. Martinotto, and G. Di Francia, 'On field calibration of an electronic nose for benzene estimation in an urban environment,' Sensors and Actuators B: Chemical, vol. 129, no. 2, pp. 750-757, 2008.

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[9] World Health Organization, 'WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide,' World Health Organization, Geneva, 2021.

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SECTION 18: DATASET LINK

The dataset utilized throughout this project is the official UCI Air Quality Benchmark Dataset, hosted by the University of California, Irvine Machine Learning Repository. In accordance with project instructions, this section serves as the final section of the report.

Dataset URL: Official UCI Repository Reference:

https://archive.ics.uci.edu/dataset/360/air+quality

Citation: De Vito, S., Massera, E., Piga, M., Martinotto, L., & Di Francia, G. (2008). Air Quality [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C59K5F.

Verification Note: The dataset was acquired directly from the UCI repository server, sanitized for sensor errors (-200 replaced with NaN), and verified with 9,357 hourly sensor instances. No additional sections or appendices follow this section.